

BASICS OF API TESTING

What is an API?

- An API (Application Programming Interface) allows communication between software components. It defines rules and protocols for accessing a web-based software application.

What is API Testing?

- It is a type of software testing that validates APIs directly to ensure they return correct responses for various requests and inputs.

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What are the types of APIs?

- Open APIs (Public)
- Partner APIs (Restricted)
- Internal APIs (Private)
- Composite APIs (Combine multiple APIs)

What protocols are used in API testing?

- HTTP, HTTPS, SOAP, REST, GraphQL, WebSocket, JMS, etc.



BASICS OF API TESTING

What are RESTful APIs?

- APIs that follow REST (Representational State Transfer) principles using standard HTTP methods like GET, POST, PUT, DELETE.

What are HTTP methods?

- GET - Retrieve data
- POST - Create data
- PUT - Update data
- DELETE - Delete data
- PATCH - Partially update data

What is an HTTP status code?

It shows the result of the request:

- 200: OK
- 201: Created
- 400: Bad Request
- 401: Unauthorized
- 404: Not Found
- 500: Server Error



BASICS OF API TESTING

What are headers in API?

- Metadata passed in API requests/responses (e.g., Content-Type, Authorization).

What is payload in an API?

- The data body sent in POST/PUT requests.

What tools are used for API testing?

- Postman
- REST Assured
- SoapUI
- JMeter
- Newman
- Swagger

What is schema validation?

- Ensuring the API response follows a defined structure (JSON Schema, XML Schema).



BASICS OF API TESTING

What is a base URI in API testing?

- The root address of the API. Example: `https://api.example.com/`.

What is an endpoint in API testing?

- A specific URL that performs a function (e.g., `/login`, `/users`).

What is query parameter vs path parameter?

- Path: `/users/{id}`
- Query: `/users?id=123`
- What is JSON and why is it used?
- JavaScript Object Notation — used for sending structured data due to its readability and compatibility.

What is response time in API testing?

Time taken by the server to respond to a request.



BASICS OF API TESTING

What is rate limiting?

- Restriction on number of API calls allowed in a specific timeframe.

What is API throttling?

- Controlling traffic to avoid overloading the server.

What are negative test cases in API testing?

- Invalid method
- Missing headers
- Invalid input
- Unauthorized access



POSTMAN & COLLECTIONS

What is Postman?

- A GUI-based tool used to test APIs with support for automation and scripting.

What is a Postman Collection?

- A group of saved requests that can be executed manually or via automation.

What is the Collection Runner?

Tool to run multiple API requests in a sequence with data files (CSV/JSON).

What is an environment in Postman?

- A set of key-value pairs (e.g., baseURL) used for dynamic test setup.

What are global variables in Postman?

- Variables accessible from any workspace or collection.



POSTMAN & COLLECTIONS

How to write test scripts in Postman?

- Using JavaScript in the Tests tab:
- Example:

```
pm.test("Status code is 200", function () {  
    pm.response.to.have.status(200);  
});
```

How to use pre-request scripts in Postman?

- Run JavaScript before sending a request (e.g., generate token).

How to test response body in Postman?

- Sample Javascript code:

```
pm.test("Name is John", function () {  
    var jsonData = pm.response.json();  
    pm.expect(jsonData.name).to.eql("John");  
});
```



POSTMAN & COLLECTIONS

How to set dynamic variables in Postman?

- Sample Javascript code:

```
pm.environment.set("userId", jsonData.id);
```

How to pass data from CSV in Postman?

- Use {{variableName}} inside the request and link the CSV in Collection Runner.

What is Newman?

- CLI companion of Postman used to run collections via command line.

How to run Newman command?

- Sample bash code:

```
newman run collection.json -e environment.json
```

Can we integrate Postman/Newman with CI tools?

- Yes, using Jenkins, GitLab CI, Azure DevOps pipelines.



POSTMAN & COLLECTIONS

What is a monitor in Postman?

- A scheduled collection run used to monitor API performance and correctness.

How to validate response time in Postman?

- Sample Javascript code:

```
pm.test("Response time < 1000ms", function ()  
{  
  pm.expect(pm.response.responseTime).to.be.below(  
    1000);  
});
```

Can we chain API calls in Postman?

- Yes, by storing response data into variables and using them in subsequent requests.

How to handle authentication in Postman?

- Options:
- Bearer Token
- OAuth 2.0
- Basic Auth
- API Keys



POSTMAN & COLLECTIONS

What is dynamic data testing in Postman?

- Using random values (e.g., email, ID) using faker.js or Postman's dynamic variables.

Can Postman validate JSON schema?

- Yes, using the tv4 or ajv library in test scripts.

What is Postman sandbox?

- The JavaScript execution environment where Postman scripts run.



REAL-TIME SCENARIOS, ERRORS & VALIDATIONS

How do you validate response data in API testing?

- Check status code
- Validate headers
- Check response body content
- Verify schema
- Response time check

What is the difference between 401 and 403?

- 401 Unauthorized: Authentication failed
- 403 Forbidden: Authenticated but access denied

How do you handle token expiration in API tests?

- Use Pre-request Script to refresh token before each request
- Store updated token in environment variables

What is a mock server in API testing?

- A fake server that mimics real API responses for testing when backend is not ready.

